

DVD RENTAL STORE – EXPLORATORY DATA ANALYSIS

PROJECT OVERVIEW

Objective

This project aims to perform exploratory data analysis (EDA) on a DVD rental database to extract business insights on customer behaviour, film popularity, revenue generation, operational efficiency, and time-based rental patterns.

The objective is not only to describe patterns, but to translate raw data into actionable business decisions capable of improving revenue efficiency, customer retention, inventory utilization, and operational performance.

Dataset

The analysis is based on the educational MySQL Sakila DVD rental database, which includes:

- Films and Categories
- Customers
- Rentals and Payments
- Staff and Stores
- Inventory

Methodology

The main analysis was conducted using SQL in MySQL Workbench and follows a structured approach:

- Data exploration using JOIN operations across relational tables
- Aggregation using COUNT, SUM, and GROUP BY
- Time-based analysis using date functions
- Customer segmentation using CASE logic and CTEs
- Revenue analysis across multiple business dimensions
- Risk analysis using late return behaviour

DATA VALIDATION

Before conducting the analysis, data quality checks were done to validate the integrity and consistency of the Sakila database. Validation included checks for missing values, duplicate records, invalid payment amounts, orphaned relationships between tables, and inconsistent date sequences (e.g., return dates preceding rental dates).

The dataset had strong structural integrity, with no significant anomalies detected in critical business tables such as rentals, payments, customers, inventory, and films. This ensured that the following analysis and business insights were based on reliable and consistent data.

1. FILM POPULARITY ANALYSIS

Objective

Analyse rental behaviour to identify the most popular films and categories in order to understand customer preferences.

Approach

- Joined rental, inventory, and film tables to calculate film-level rental frequency
- Extended analysis using film_category and category tables
- Aggregated rental counts using COUNT grouped by film and category

Key Findings

Film-Level Insights

- Top rented films are *Bucket Brotherhood*, *Rocketeer Mother*, and *Zorro Ark*
- Top rental counts are tightly clustered (31–34 rentals)
- No single film dominates demand

Category-Level Insights

- Most popular categories: *Sports*, *Animation*, *Action*, *Sci-Fi*
- Category demand is evenly distributed across genres

Insights

- Demand varies greatly across the catalogue
- Customers show preference for genre variety rather than specific blockbuster titles
- Category preference seems to be the primary driver of demand, suggesting that inventory stocking should be optimized at the genre level rather than individual film level.

Business Recommendations

- Maintain a diverse film catalogue rather than focusing on a few titles
- Prioritize high-performing categories such as Sports, Action, and Sci-Fi
- Prioritize inventory allocation and licensing budget toward high-performing categories rather than individual titles
- Reduce risk of underutilized inventory by avoiding overinvestment in single-film promotion strategies

2. CUSTOMER SEGMENTATION ANALYSIS

Objective

Classify customers based on engagement and revenue contribution. Later in our analysis, customers will also be assessed in their reliability (late return behaviour), as we aim to see if there is a possible negative correlation, in spite of their profitability.

Approach

- Calculated rental frequency per customer
- Calculated total revenue per customer
- Combined metrics using CTEs
- Segmented customers using rule-based CASE classification

Key Findings

- High-value customers consistently show both high rental frequency and high spending
- Medium and low-value segments follow expected decline in contribution
- Strong correlation exists between engagement and revenue

Insights

Customer value is strongly correlated with rental activity, meaning engagement directly drives revenue growth.

High-value customers represent a disproportionately large share of total revenue, making them critical to business sustainability.

Business Recommendations

- Focus retention strategies on high-value customers
- Encourage medium-value customers to increase rental frequency
- Introduce tier-based loyalty programs targeting medium-value customers to increase lifetime value
- Prioritize retention of high-value customers, over acquisition, as they provide the highest marginal return of investment (ROI)

3. LATE RETURNS & CUSTOMER RISK ANALYSIS

Objective

Assess late returns and estimate potential revenue loss due to late fees. Classify customers in their return behaviour in order to compare with their profitability.

Approach

- Calculated expected return date using rental duration
- Compared expected vs actual return dates
- Classified returns into *On Time*, *Late* and *Not Returned*
- Calculated: *Late days*, *Late fee (figurative 0.50€ per day)*, *Late frequency per customer*

Key Findings

Customer Behaviour

- High-value customers often exhibit high lateness rates.
- Late rates range from ~40% to over 65% for some customers
- Lateness is present across all segments, not isolated cases

Revenue Impact

- Late fees generate consistent additional revenue per customer
- High-value customers contribute disproportionately to both total revenue and late return activity, creating an operational trade-off between customer retention and inventory efficiency.

Insights

- High-value customers are responsible for a significant share of both revenue and late returns, indicating that revenue generation and operational reliability are not aligned.
- Late return behaviour is systemic rather than isolated, with most customers exhibiting moderate lateness rates (~45–60%).
- This suggests that current rental policies may be misaligned with customer behaviour.

Business Recommendations

- Re-evaluate rental duration policies
- Consider increasing late fee rates.
- Instead of applying uniform penalties, implement a segmented policy (avoid revenue loss from top customers while still improving operational efficiency):
 - High-value customers → soft enforcement (reminders, grace periods)
 - Low-value / high-risk customers → stricter penalties

4. RENTAL TRENDS OVER TIME

Objective

Analyse rental and revenue patterns over time, in monthly and daily level.

Approach

- Extracted month and weekday from rental dates
- Aggregated rental volume and revenue per time period
- Joined rental and payment tables for combined analysis

Key Findings

Monthly Trends

- Rentals increase steadily over time
- Peak activity occurs in mid-year (July–August)
- Early months show significantly lower activity

Weekly Trends

- Highest activity occurs on Tuesday
- Weekend activity is strong but not dominant
- Demand is stable across all days

Insights

- Demand stability indicates predictable operational load, reducing the need for additional staff on specific weekdays.
- Peak activity in mid-year/summer suggests seasonal demand cycles, likely tied to external factors (e.g., holidays or leisure periods)

Business Recommendations

- Maintain consistent staffing throughout the week
- Optimize staffing based on stable demand rather than an over-allocating strategy on weekends.
- Prepare inventory scaling for mid-year demand spikes
- Align marketing campaigns with peak seasonal demand to maximize profit relative to investment.

5. CATEGORY PROFITABILITY ANALYSIS

Objective

Identify which film categories generate the highest revenue and profitability. The analysis below complements the finding of analysis #1, in finding the most profitable categories.

Approach

- Merged rental data with payment data per rental
- Joined with category table
- Aggregated revenue and rental count per category
- Calculated average revenue per rental

Key Findings

Revenue Leaders

Sports, Sci-Fi, Animation, Drama

Demand Leaders

Sports, Animation, Action, Sci-Fi

Insights

Demand $\neq$ Profitability

Some categories generate high rental volume but lower revenue per rental, while others generate higher revenue per rental despite lower demand.

Category performance should be evaluated using both volume (demand) and value (revenue per rental), because optimizing for only one dimension leads to less optimal business outcomes.

High-value categories (avg revenue per rental)

Comedy, New, Sports, Games, Horror

Low-value categories

Family, Children, Classics, Action

Business Recommendations

- Prioritize high-value categories for profitability
- Maintain high-volume categories (*e.g., Animation, Action*) for customer retention
- Optimize pricing strategy for low-value categories
- Balance inventory between demand-driven categories and profit-driven categories

6. STAFF PERFORMANCE ANALYSIS

Objective

Evaluate staff performance based on revenue and efficiency. Unfortunately the data wasn't sufficient for a more meaningful analysis (only 2 distinct employees in our Sakila DB).

Approach

- Joined rental, payment, and staff tables
- Calculated: *Total transactions*, *Total revenue*, *Revenue per rental (efficiency)*

Key Findings

- Both staff members handle nearly identical workloads (~8,000 transactions)
- Revenue generation is nearly balanced across staff
- Small efficiency difference exists:
 - *Jon Stephens*: slightly higher revenue per rental
 - *Mike Hillyer*: slightly lower efficiency

Insights

The minimal variance between staff performance metrics suggests a standardized operational process and consistent service execution across employees. Minor efficiency variation suggesting differences in transaction quality or customer handling.

Business Recommendations

- Maintain current staffing structure
- Investigate small efficiency differences for optimization
- Use efficiency metrics for performance benchmarking for future staff evaluation
- Ensure training consistency across staff

7. STORE PERFORMANCE ANALYSIS

Objective

Evaluate store-level performance in terms of rentals and revenue. Unfortunately the data wasn't sufficient for a more meaningful analysis (only 2 distinct stores in our Sakila DB).

Key Findings

- Store performance is almost perfectly balanced
- Revenue split:
 - *Store 1: 49.73%*
 - *Store 2: 50.27%*
- Slight efficiency advantage in Store 2

Insights

The business operates with highly balanced store performance, indicating consistent demand distribution and operational stability. The slight efficiency advantage in Store 2 could be a potential for best practice replication across stores. However, the near-equal revenue distribution (~50/50 split) suggests strong geographic balance and low concentration risk between store locations.

Business Recommendations

- Maintain equal resource allocation across stores
- Use store-level performance as a stability benchmark and baseline for scalability, ensuring consistency in future expansion.
- Monitor small efficiency differences for optimization
- Avoid unnecessary restructuring on store level

SUMMARY

This analysis reveals a highly balanced and stable DVD rental business characterized by the following findings:

- High-value customers drive a disproportionate share of total revenue, but also exhibit elevated late-return behaviour, creating a trade-off between customer retention and operational efficiency.
- Category-level demand is more influential than individual film performance, suggesting inventory strategy should prioritize genre optimization rather than blockbuster dependency.
- Revenue and operational activity are highly balanced across staff and stores, indicating stable business execution and low operational concentration risk.
- Stable weekly demand patterns suggest opportunities for cost optimization through balanced staffing and inventory allocation.

OVERALL BUSINESS INSIGHTS

The business demonstrates a stable and diversified revenue structure driven by broad customer engagement across multiple film categories rather than dependence on isolated high-performing products.

However, operational inefficiencies (particularly late return behaviour) highlight a key opportunity for management to improve profitability not through expansion, but through operational optimization and customer policy refinement.

The highest-impact opportunity lies in combining customer behaviour with operational efficiency, through:

- customer-level segmentation strategies
- data-driven rental policy optimization
- data-driven pricing and promotions